

Name: Aashray Lal Kayastha

PRN:250770122005

Division: A1

Batch:2025-2029

Sem:3

Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Outcome:

1. Default Constructor:

Source code:

```
1  class student{
5  student(){
7      age=18;
8  }
9
10 void display(){
11     System.out.println("Name:"+name+",Age:"+age);
12 }
13 }
14
15 public class exp2_outcome2{
    Run main | Debug main | Run | Debug
16     public static void main(String[] args) {
17         student student1=new student();
18         student1.display();
19     }
20 }
```

Output:

```
c:\Users\Aashray\Documents\University\3rd sem\JAVA>java exp2_outcome2
Name:Aashray Kayastha,Age:18
```

2. Parametrized Constructor:

Source code:

```
1  class Student{
2      string name;
3      int age;
4
5      Student(String n,int a){
6          name=n;
7          age=a;
8      }
9
10     void display(){
11         System.out.println("Name:"+name+",Age:"+age);
12     }
13 }
14
15
16 public class exp2_outcome1{
17     Run main | Debug main | Run | Debug
18     public static void main(String[] args) {
19         Student student1=new Student(n: "Aashray",a: 005);
20         student1.display();
21     }
22 }
```

Output:

```
c:\Users\Aashray\Documents\University\3rd sem\JAVA>java exp2_outcome1.java
Name:Aashray,Age:5
```

3) Copy Constructor

Source Code:

```
1  class Student {
2      String name;
3      int age;
4
5      // Parameterized Constructor
6      Student(String n, int a) {
7          name = n;
8          age = a;
9      }
10
11     // Copy Constructor
12     Student(Student s) {
13         name = s.name;
14         age = s.age;
15     }
16
17     void display() {
18         System.out.println("Name: " + name + ", Age: " + age);
19     }
20 }
21
22 public class demo {
23     public static void main(String[] args) {
24         Student student1 = new Student("David", 20);
25         Student student2 = new Student(student1);
26
27         student1.display();
28         student2.display();
29     }
30 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java demo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Name: David, Age: 20
Name: David, Age: 20
```

Conclusion:

This program demonstrates the use of constructors in Java through the Student class. The default constructor assigns fixed values, the parameterized constructor allows custom student data, and the copy constructor creates a new object using an existing student's values. Together, they show how object initialization can be handled flexibly in Java.

Exercise

1. Create a **student class** using default and parameterized constructors to initialize student name and roll number.

Source code:

```
1  class Student{
2      String name;
3      int roll;
4
5      Student() {
6          name = "Unknown";
7          roll = 0;
8      }
9
10     Student(String n, int r) {
11         name = n;
12         roll = r;
13     }
14
15     void display() {
16         System.out.println("Name = " + name + ", Roll No = " + roll);
17     }
18 }
19
20 public class exp2_a{
21     Run main | Debug main | Run | Debug
22     public static void main(String[] args) {
23         Student student1 = new Student(n: "Aashray", r: 20);
24         student1.display();
25
26         Student student2 = new Student();
27         student2.display();
28     }
29 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA> cmd /C ""C:\Program Files\Java\jdk-26.0.1\bin\java.exe" --enable-preview -XX:+
ShowCodeDetailsInExceptionMessages -cp C:\Users\Aashray\AppData\Roaming\Code\User\workspaceStorage\3edaa2e7651374d9c42764f7255b333
3\redhat.java\jdt_ws\JAVA_1fb59382\bin exp2_a "
Name = Aashray, Roll No = 20
Name = Unknown, Roll No = 0
```

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

Source code:

```
1  //2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile
2  //details and create duplicate object records.
3  class Mobile{
4      String brand;
5      int price;
6
7      Mobile(){
8          brand="Apple";
9          price=60000;
10     }
11
12     Mobile(String b,int p){
13         brand=b;
14         price=p;
15     }
16
17     Mobile(Mobile m){
18         brand=m.brand;
19         price=m.price;
20     }
21
22     void display(){
23         System.out.println("Brand Name="+brand+"    Price="+price);
24     }
25
26 }
27
28 public class exp2_b {
29     Run main | Debug main | Run | Debug
30     public static void main(String[] args) {
31         Mobile mob1=new Mobile(b: "Oneplus",p: 50000);
32         mob1.display();
33         Mobile mob2=new Mobile();
34         mob2.display();
35         System.out.println(x: "Duplicate Object Record:");
36         Mobile mob3=new Mobile(mob1);
37         mob3.display();
38     }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA> cmd /C ""C:\Program Files\Java\jdk-26.0.1\bin\java.exe" --enable-preview -XX:+
ShowCodeDetailsInExceptionMessages -cp C:\Users\Aashray\AppData\Roaming\Code\User\workspaceStorage\3edaa2e7651374d9c42764f7255b333
3\redhat.java\jdt_ws\JAVA_1fb59382\bin exp2_b "
Brand Name=Oneplus    Price=50000
Brand Name=Apple    Price=60000
Duplicate Object Record:
Brand Name=Oneplus    Price=50000
Activate Windows
```